

Ricardo Castro Figueiredo

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Experience

Critical Manufacturing

Maia, Portugal

Software Developer - Intern

July 2024 – Sept. 2024

- Designed and deployed Feedback Circle, a full-stack web application that streamlined the company's performance appraisal process
- Implemented a component-based architecture using vanilla JavaScript, HTML, and CSS for the frontend, with Node.js and MySQL for the backend
- Designed and built a RESTful API handling user authentication, feedback management, and role-based visibility control
- Created a database schema to manage feedback workflows between employees, managers, and appraisers

Projects

- **Kotlin Compiler** *Haskell, Alex/Happy, MIPS*
 - Developed compiler with lexer/parser (Alex/Happy), semantic checks, and MIPS codegen for Kotlin constructs
 - Optimized via graph coloring register allocation and liveness analysis
- **CART Class Imbalance** *Python, scikit-learn, SMOTE*
 - Improved Recall score by 18% with weighted Gini impurity for minority classes
 - Validated via stratified sampling and ROC-AUC metrics across several datasets
- **Roomba Battle Bot** *Java, Robocode*
 - Predictive targeting using trigonometry and adaptive bullet power
 - Energy-efficient evasion via collision timeout tracking + zigzag movement
- **Shinsu Duel: a Tower of God CCG** *Node.js, Socket.IO, Express*
 - Real-time PvP/PvE card game with synced state management
 - Component-based UI with drag-and-drop interactions
- **Cryptography Benchmark** *Python, Cryptography, Matplotlib*
 - Automated performance analysis of AES/RSA/SHA-256
 - Visualized encryption-decryption trends with statistical averaging

Education

Bachelor in Computer Science

University of Porto

Expected Feb. 2026

- **Relevant Coursework:** Algorithms & Data Structures, AI/ML, Computer Systems, Databases & Big Data, Security & Cryptography, Software Architecture, Mathematics, Programming Paradigms, Theory of Computation, Human-Machine Interaction

Skills

- **Languages:** Python, Java, C++, Haskell, SQL, JavaScript, HTML/CSS, Bash/Shell, LaTeX
- **Tools & Frameworks:** Pandas, NumPy, MySQL, MongoDB, Node.js, Express.js, REST APIs, TCP/IP, AMPL, Jupyter, Figma, Git, React
- **Platforms:** AWS, GCP, Linux/Unix, Docker, Anaconda

Certifications

- English C1 Advanced (CAE), Cambridge Assessment English *June 2021*